CHUONG 3: CAY NHI PHAN & CAY NHI PHAN TIM KIEM (BST)

Mon hoc: Cau truc du lieu va giai thuat (INT1306) - PTIT

1. Dinh nghia Cay nhi phan (Binary Tree):
- Cay nhi phan la cau truc du lieu phan cap, trong do moi nut (Node) co toi da 2 nut con: nut con trai (left child) va nut con phai (right child).
- Cau truc Node trong C++:

```
struct Node {
    int data;
    Node* left;
    Node* right;
};
```

2. Tinh chat Cay nhi phan tim kiem (Binary Search Tree - BST):
- Moi nut thuoc cay con ben trai co gia tri NHO HON nut goc (left->data < root->data).
- Moi nut thuoc cay con ben phai co gia tri LON HON nut goc (right->data > root->data).
- Hai cay con trai va phai cung phai la cay nhi phan tim kiem.

CAC PHEP DUYET CAY NHI PHAN (TREE TRAVERSALS)

Ba phep duyet cay de quy kinh dien (Do phuc tap O(n)):

1. Duyet Tien thu tu - Pre-order (Node - Left - Right):

```
void preOrder(Node* root) {
    if (root == nullptr) return; // B?T BU?C: ?i?u ki?n d?ng ?? quy!
    cout << root->data << ' ';
    preOrder(root->left);
    preOrder(root->right);
}
```

2. Duyet Trung thu tu - In-order (Left - Node - Right):
- Tren cay BST, phep duyet In-order se in ra cac phan tu theo thu tu TANG DAN.

3. Duyet Hau thu tu - Post-order (Left - Right - Node):
- Ung dung: Tinh chieu cao cay, giai phong bo nho (delete tree).

TIM KIEM, THEM NUT VA XOA NUT TRONG BST

1. Tim kiem (Search) & Them nut (Insert):
- Do phuc tap trung binh: O(log n) khi cay can bang.
- Truong hop xau nhat (cay suy bien thanh danh sach lien ket): O(n).

2. Xoa mot nut (Delete Node) trong BST gom 3 truong hop:
- TH1: Nut can xoa la nut la (khong co con) -> Giai phong truc tiep.
- TH2: Nut can xoa co 1 cay con -> Noi cha cua nut do voi nut con duy nhat.
- TH3: Nut can xoa co du 2 con -> Tim nut the mang la phan tu NHO NHAT ben cay con phai (In-order Successor), chep gia tri vao nut can xoa roi xoa ?? quy nut the mang.